

1. Introduction

In today's rapidly evolving digital infrastructure landscape, real-time system monitoring has become a critical requirement for maintaining the health, performance, and reliability of computing environments. Traditional monitoring tools often rely on manual inspection or simple threshold-based alerts, which are reactive in nature and insufficient for detecting subtle performance degradations before they escalate into critical failures. To address these limitations, CloudPulse is designed as an AI-powered, real-time system monitoring platform that combines live metric collection, intelligent anomaly detection, and an interactive web dashboard into a unified solution.

CloudPulse continuously collects key system resource metrics including CPU utilization, RAM consumption, disk usage, network I/O, and active process statistics — from the host machine using a lightweight Python monitoring agent built on the psutil library. These metrics are transmitted every five seconds to a FastAPI backend, where they are persisted in a PostgreSQL database, cached in Redis for sub-millisecond reads, and analyzed in real time using a machine learning model based on the Isolation Forest algorithm. When anomalous behavior is detected or predefined thresholds are breached, the system automatically generates alerts and broadcasts live updates to all connected dashboard clients through a WebSocket connection. The frontend, built with React 18 and Vite, presents this data through an elegant dark-themed interface featuring live charts, metric cards, anomaly feeds, and alert panels — enabling operators to gain immediate, actionable insights into system health without any manual intervention.

This project demonstrates the practical integration of modern full-stack web technologies, asynchronous programming paradigms, real-time communication protocols, and unsupervised machine learning in a single cohesive application. The system is fully containerized using Docker Compose, making it straightforward to deploy across different environments with a single command.

1.1 Technology Stack

The development of CloudPulse required a carefully selected set of modern technologies, each chosen to fulfill a specific role within the system's architecture. At the monitoring layer, the Python library `psutil` is used to extract real-time operating system metrics such as CPU frequency, memory usage, disk I/O counters, and network throughput directly from the host machine. The monitoring agent communicates with the backend over HTTP using `httpx`, an asynchronous HTTP client that supports retry logic without blocking the collection loop.

The AI/ML component is powered by `scikit-learn`'s `IsolationForest` algorithm, combined with `StandardScaler` for feature normalization. The model operates on seven features — five raw metrics and two engineered features (`cpu_ram_pressure` and `net_total_mb`) — to detect multi-dimensional anomalies that would be invisible when examining individual metrics in isolation. Model training is performed in a background thread pool using `asyncio`'s `run_in_executor`, ensuring the CPU-bound `sklearn` computation never blocks the event loop.

On the frontend, the user interface is built with `React 18` and bundled with `Vite` for fast hot-module replacement during development. `Recharts` is used for all data visualizations including area charts, bar charts, and sparklines, while `Tailwind CSS` provides the utility-first styling system with a custom dark theme, glassmorphism card effects, and micro-animation keyframes. Navigation is handled by `React Router v6`, and all REST API communication is performed through `Axios` with a global response interceptor for error logging. The `Lucide React` icon library is used throughout the interface for consistent, lightweight iconography. Finally, the entire application stack — including `PostgreSQL`, `Redis`, the `FastAPI` backend, the monitoring agent, and the `React` frontend — is containerized using `Docker` and orchestrated with a single `docker-compose.yml` file, enabling one-command full-stack deployment.

1.2 Scope of the Project

The scope of CloudPulse encompasses the design, development, and deployment of a complete end-to-end system monitoring solution that operates in real time. The project covers the full software lifecycle — from low-level operating system metric collection on the host machine, through backend data processing and persistence, to a fully interactive browser-based dashboard consumed by end users. The system is scoped to monitor a single host machine in its primary implementation, with the architectural foundation laid to support multi-host environments in future iterations through hostname-based filtering already present in the data model and query layer.

Within the scope of metric collection, CloudPulse tracks five core resource categories: CPU utilization and frequency, RAM usage in both percentage and gigabytes, disk storage consumption and I/O read/write throughput, network inbound and outbound bandwidth in megabytes, and a live snapshot of the top ten processes sorted by CPU consumption. These metrics are collected at a fixed interval of every five seconds, balancing granularity with system overhead.

The project also encompasses an AI-powered anomaly detection pipeline that automatically trains an Isolation Forest model on historical data and applies it to every incoming metric snapshot. This includes feature engineering, per-feature contribution scoring, severity classification, and persistent storage of detected anomalies. A rule-based alerting engine runs in parallel, generating threshold alerts when CPU exceeds 85%, RAM exceeds 85%, or disk usage surpasses 90%, with built-in deduplication logic to prevent alert flooding within five-minute windows.

On the presentation layer, the scope includes four distinct pages — Dashboard, Analytics, Alerts, and Processes — each serving a different operational need. The Dashboard provides live situational awareness, Analytics enables historical trend exploration across configurable time windows, Alerts supports alert lifecycle management with filtering and resolution workflows, and Processes offers a searchable, sortable view of running system processes. Real-time data delivery is handled through a persistent WebSocket connection with exponential-backoff reconnection, heartbeat management, and concurrent fan-out broadcasting.

The project is explicitly scoped to function as a learning and demonstration platform, and therefore does not include features such as multi-user authentication, role-based access control, cloud infrastructure monitoring (e.g., AWS/GCP/Azure APIs), mobile application support, or integration with external notification channels such as email or SMS. These capabilities represent natural extensions that fall outside the current project boundary but have been considered in the system's modular architecture to facilitate future growth.

2. Motivation

The motivation behind the development of CloudPulse stems from a fundamental challenge faced by developers, system administrators, and DevOps engineers alike — the lack of an accessible, intelligent, and real-time window into the health of their computing systems. While enterprise-grade monitoring tools such as Grafana, Prometheus, and Datadog exist, they are often complex to configure, resource-intensive to run, and require significant prior infrastructure knowledge. For students and learners seeking to understand how modern monitoring platforms operate at a technical level, these tools serve as black boxes rather than educational references. CloudPulse was conceived to bridge this gap — to build a monitoring system from first principles, where every component from data collection to

anomaly detection to live visualization is transparent, understandable, and purpose-built.

A second and equally compelling motivation is the growing importance of proactive system management over reactive incident response. In many real-world scenarios, system failures — such as memory leaks, runaway processes, or disk exhaustion — do not occur instantaneously. They manifest gradually as subtle statistical deviations in resource usage patterns before crossing critical thresholds. Traditional alerting systems, which fire only when a hard threshold is breached, miss this early warning window entirely. The integration of unsupervised machine learning, specifically the Isolation Forest algorithm, into CloudPulse was motivated by the desire to detect these anomalies at an early stage — before they cause service disruption — and to do so without requiring labeled training data, which is rarely available in operational environments.

The motivation also extends to the domain of software engineering education. Building CloudPulse required the convergence of multiple advanced concepts — asynchronous programming in Python, WebSocket-based real-time communication, Redis pub/sub architecture for horizontal scalability, PostgreSQL time-series query optimization, React context-driven state management, and scikit-learn model lifecycle management including serialization, background retraining, and inference. Each of these components presents a real-world engineering challenge that cannot be fully appreciated through textbook study alone. By implementing them in a cohesive, working system, this project serves as a practical demonstration of how modern backend and frontend technologies can be composed together to solve a meaningful problem.

Finally, the project was motivated by the vision of creating something visually compelling and immediately useful. A monitoring dashboard that is difficult to read or aesthetically uninspiring discourages adoption. CloudPulse was designed with a premium dark-themed interface, smooth animations, glassmorphism card effects, live sparklines, and color-coded severity indicators — not merely for

aesthetics, but because visual clarity directly impacts how quickly an operator can assess system health and respond to issues. The belief that good engineering and good design are not mutually exclusive was a driving principle throughout the development of this project.

3. Related Work

System monitoring and anomaly detection are well-established research and engineering domains with a rich body of existing work spanning both academic literature and industry tooling. This section reviews the most relevant existing solutions, their capabilities, and how CloudPulse differentiates itself in terms of design philosophy, technical approach, and target use case.

3.1 Existing Monitoring Platforms

Prometheus is one of the most widely adopted open-source monitoring systems, originally developed at SoundCloud and later donated to the Cloud Native Computing Foundation (CNCF). Prometheus operates on a pull-based metric collection model, where the server periodically scrapes HTTP endpoints exposed by instrumented applications. While extremely powerful and scalable, Prometheus requires significant configuration overhead — defining scrape intervals, writing PromQL queries, and setting up exporters for each monitored resource. It does not natively provide anomaly detection, and its visualization layer depends on an external tool, Grafana, which must be separately installed and configured. CloudPulse, by contrast, uses a push-based agent model and integrates collection,

storage, AI analysis, and visualization into a single cohesive application requiring zero external configuration.

Datadog is a leading commercial observability platform offering infrastructure monitoring, application performance management (APM), log aggregation, and AI-powered anomaly detection through its Watchdog feature. Datadog's anomaly detection relies on proprietary seasonal decomposition algorithms trained on cloud-scale datasets. While highly capable, Datadog is a closed-source, subscription-based SaaS product that provides no visibility into its internal algorithms or data processing pipelines. CloudPulse was motivated in part by the desire to implement a transparent, open, and explainable anomaly detection mechanism — specifically Isolation Forest with per-feature contribution scores — so that every detection decision can be understood and audited by the operator.

Grafana + InfluxDB represents another popular open-source stack, where InfluxDB serves as a purpose-built time-series database and Grafana provides rich panel-based dashboards. This combination offers excellent scalability for high-cardinality metric streams and supports alerting rules through Grafana's unified alerting system. However, the stack requires the user to configure and maintain three or more separate services (collector, database, dashboard), write Flux or InfluxQL queries, and manually design dashboard panels. Machine learning capabilities require additional integration with tools like MLflow or custom Grafana plugins. CloudPulse consolidates all of these concerns into a single deployable stack with a purpose-built AI pipeline.

Netdata is a lightweight, open-source real-time monitoring agent that runs on the host and streams metrics to a web dashboard at one-second resolution. It offers an impressive out-of-the-box experience with automatic anomaly detection using streaming machine learning models per metric. However, Netdata's ML approach applies univariate anomaly detection independently to each metric stream, meaning it cannot detect anomalies that only manifest when multiple resources are

simultaneously stressed — for example, moderate CPU combined with moderate RAM and a network spike. CloudPulse's Isolation Forest operates on all features simultaneously as a multivariate model, capturing these combined-load anomalies that univariate approaches miss.

3.2 Anomaly Detection Research

The use of Isolation Forest for system metric anomaly detection has been explored in several academic and industrial contexts. Liu et al. (2008) introduced the Isolation Forest algorithm, demonstrating its effectiveness for anomaly detection in high-dimensional datasets without requiring density estimation or distance computation, making it significantly faster than traditional approaches such as Local Outlier Factor (LOF) or One-Class SVM. Subsequent work by Emmott et al. (2015) benchmarked anomaly detection algorithms across diverse datasets, finding that Isolation Forest consistently performs well in low-to-medium dimensional settings — aligning with CloudPulse's seven-feature input space.

In the context of IT operations, AIOps (Artificial Intelligence for IT Operations) has emerged as a research domain that applies machine learning to log analysis, root cause identification, and predictive alerting. Research by Dang et al. (2019) at Microsoft demonstrated that unsupervised anomaly detection methods, including Isolation Forest variants, can effectively identify performance degradations in distributed systems before they trigger traditional threshold-based alerts. CloudPulse draws directly from this philosophy, applying unsupervised detection at the single-host level as a practical, lightweight implementation of AIOps principles.

3.3 Real-Time Dashboard Architecture

The architectural pattern of combining a WebSocket-based live feed with a REST API for historical data has been validated by numerous production systems. The

use of Redis Pub/Sub as a broadcast intermediary — decoupling metric ingestion from WebSocket delivery — mirrors patterns employed in large-scale systems such as Slack's real-time messaging infrastructure and Uber's surge pricing dashboards, as documented in their respective engineering blogs. CloudPulse adopts this pattern at a smaller scale to ensure architectural correctness and horizontal scalability, even within a single-instance learning project.

3.4 Positioning of CloudPulse

Compared to the tools and research discussed above, CloudPulse occupies a unique position as a self-contained, educationally transparent, full-stack monitoring system that integrates real-time collection, persistent storage, explainable AI anomaly detection, rule-based alerting, and a premium interactive dashboard — all within a single Docker Compose deployment. It does not aim to replace enterprise-grade platforms but rather to demonstrate, in a pedagogically valuable way, how each layer of a modern monitoring system is designed and interconnected. The explainability of its anomaly detection — through per-feature z-scores and contribution weights — represents a conscious design choice that distinguishes it from black-box commercial solutions.

4. Feasibility Study

4.1 Technical Feasibility

The technical feasibility of CloudPulse was assessed by evaluating whether the required technologies, algorithms, and integration patterns could be implemented reliably within the constraints of a development environment running on a standard

personal computer. The assessment concluded that the project is fully technically feasible, and the completed implementation serves as direct evidence of this.

From a hardware perspective, CloudPulse imposes minimal resource requirements. The monitoring agent itself is a lightweight Python process that uses less than 30 MB of RAM and negligible CPU, as psutil makes direct system calls rather than spawning subprocesses. The FastAPI backend, PostgreSQL database, and Redis cache together operate comfortably within 512 MB of RAM during normal load, making the entire stack deployable on any modern laptop or low-cost cloud virtual machine. The frontend is a statically served React single-page application that requires no server-side rendering resources.

The WebSocket architecture presented a challenge around horizontal scalability — if multiple backend instances are deployed, only the instance that received the HTTP POST from the agent would broadcast the update to its connected clients. This was resolved using Redis Pub/Sub as a message bus, where the ingest endpoint publishes to a shared channel and all backend instances subscribe and relay to their respective WebSocket clients. This pattern is technically sound and validated by industry practice, even though single-instance deployment is the current scope.

4.2 Economic Feasibility

The economic feasibility of CloudPulse is exceptionally strong. The project was developed entirely using free and open-source software, resulting in zero licensing costs across the entire technology stack. Python, FastAPI, SQLAlchemy, scikit-learn, psutil, httpx, Redis, PostgreSQL, React, Vite, Tailwind CSS, Recharts, and Docker are all available under permissive open-source licenses (MIT, Apache 2.0, BSD, or PostgreSQL License) that permit free use in both educational and commercial contexts.

Development costs are limited to the time investment of the developer and the use of a standard personal computer. No paid cloud infrastructure was required — the entire stack runs locally via Docker Compose, eliminating hosting expenses during development and testing. For production deployment, CloudPulse can be hosted on a minimal virtual machine such as an AWS EC2 t3.micro instance or a DigitalOcean Droplet at approximately \$5–\$10 USD per month, making it economically viable for small teams or individual operators.

Maintenance costs are also minimal. The application requires no paid support contracts, and the automated data retention task — which purges metric rows older than seven days every six hours — prevents unbounded database growth, keeping storage costs predictable. The model retraining loop runs automatically every hour using existing historical data, requiring no human intervention or paid ML platform subscription (such as AWS SageMaker or Google Vertex AI).

4.3 Operational Feasibility

Operational feasibility evaluates whether CloudPulse can be realistically deployed, used, and maintained by its intended operators in a day-to-day environment. The assessment confirms that the project is operationally feasible across all dimensions of usability, deployability, maintainability, and reliability.

Ease of deployment is a primary operational strength of CloudPulse. The entire four-service stack — PostgreSQL, Redis, FastAPI backend, monitoring agent, and React frontend — is containerized using Docker and orchestrated with a single `docker-compose.yml` file. An operator with Docker installed can bring the complete system online with a single command (`docker-compose up --build`), with no manual database setup, dependency installation, or environment configuration beyond copying a `.env` file. Docker health checks on the PostgreSQL and Redis

containers ensure that dependent services (backend and agent) only start after their dependencies are ready, preventing startup race conditions.

Usability of the dashboard was a central design priority. The React frontend requires no technical knowledge to operate — metrics are displayed automatically as soon as the WebSocket connection is established, alerts appear in real time with clear severity labels and one-click resolution, the Processes page provides searchable and sortable live data, and the Analytics page allows non-technical users to explore historical trends through simple time-range pill buttons. Skeleton loaders prevent empty or jarring initial states, and toast notifications inform users of new anomalies and alerts without requiring them to actively monitor the page.

Reliability is addressed through several mechanisms. The monitoring agent implements a three-attempt retry loop with a two-second delay between retries, ensuring that transient network errors or brief backend unavailability do not result in data loss. The WebSocket client implements exponential backoff reconnection — starting at one second and capping at thirty seconds — so dashboard clients automatically recover from network interruptions without user intervention. Redis connection health is validated at startup and monitored every thirty seconds through a background keepalive ping. The FastAPI backend registers global exception handlers that return structured JSON error responses rather than crashing on unexpected inputs, and the database session manager automatically rolls back transactions on error and closes sessions in all cases through a finally block.

Maintainability is supported by the project's modular architecture. Each concern — metric collection, HTTP ingestion, alert checking, anomaly detection, WebSocket broadcasting — is implemented as a separate, independently testable service or module with clearly defined interfaces. Configuration is centralized in a single Settings class loaded from environment variables, making it straightforward to adjust thresholds, retrain intervals, or connection parameters without modifying source code. The automated data retention task runs every six hours to purge

metrics older than seven days, preventing the database from growing unboundedly over time and maintaining consistent query performance.

5. Methodology / Planning of Work

The development of CloudPulse followed an iterative, bottom-up development methodology, where each layer of the system was built, tested, and validated independently before being integrated with the next. This approach was chosen over a waterfall model because the interdependencies between components — particularly between the backend API, the WebSocket broadcasting system, and the React dashboard — required frequent feedback loops to ensure compatibility and correctness. The methodology can be broadly described in five sequential phases.

5.1 Requirements Analysis and Architecture Design

The first phase involved a thorough analysis of what a real-time AI-powered monitoring system must deliver. Functional requirements were identified across all four layers: the agent must collect system metrics at a fixed interval and transmit them reliably; the backend must ingest, persist, analyze, and broadcast metrics with minimal latency; the database must support both high-frequency inserts and time-range aggregate queries efficiently; and the frontend must display live data without perceptible lag while remaining navigable and informative. Non-functional requirements — including fault tolerance, horizontal scalability via Redis pub/sub, and a sub-10-millisecond API response time for cached reads — were also defined at this stage.

Following requirements gathering, the system architecture was designed as a four-layer pipeline: Agent → FastAPI Backend → PostgreSQL + Redis → React Dashboard. Key architectural decisions made during this phase included the choice of a push-based agent model over a pull-based scrape model (simplifying agent configuration), the use of Redis pub/sub to decouple WebSocket broadcasting from metric ingestion, the selection of Isolation Forest for its suitability to unlabeled operational data, and the decision to run model retraining in a background thread pool to preserve event loop responsiveness.

5.2 Backend Development

The second phase focused on building the FastAPI backend in a layered fashion. The database models (SystemMetric, Anomaly, Alert) and their corresponding Pydantic schemas were defined first, establishing the data contracts that all subsequent layers would depend upon. The services layer — MetricsService, AlertService, AnomalyService, and RedisService — was implemented next, encapsulating all business logic and database interaction behind clean interfaces. The API routers were then built on top of the services layer, followed by the middleware, exception handlers, and the FastAPI lifespan manager that orchestrates startup and shutdown sequences. The WebSocket manager and Redis pub/sub broadcaster were developed last, as they depended on the completed services layer.

5.3 AI/ML Pipeline Development

The third phase involved the development of the anomaly detection pipeline. The `AnomalyDetector` class was designed as a singleton that loads a pre-trained model from disk on startup and exposes a `predict()` method returning a boolean flag, anomaly score, and per-feature contribution dictionary. Feature engineering — computing `cpu_ram_pressure` and `net_total_mb` — was implemented as a pure function applied consistently during both training and inference to prevent training-serving skew. The `ModelTrainer` module was built to query the last 24 hours of metrics from PostgreSQL, convert ORM objects to plain dictionaries, and invoke `anomaly_detector.train()` in a thread pool. A sample data generator (`sample_data.py`) was created to seed the database with 500 normal and 25 anomalous records, enabling model training without waiting for real production data to accumulate.

5.4 Monitoring Agent Development

The fourth phase produced the lightweight Python monitoring agent. Each resource type (CPU, RAM, Disk, Network, Processes) was implemented as a separate collector module returning a plain dictionary, following the single-responsibility principle. The `monitor.py` main loop merges all collector outputs into a single payload and passes it to `sender.py`, which manages HTTP POST delivery with retry logic. The agent was designed to be stateless requiring only the `BACKEND_URL` and `COLLECT_INTERVAL` environment variables enabling straightforward containerization.

5.5 Frontend Development

The fifth phase built the React dashboard. Development began with the global design system — defining Tailwind color tokens, animation keyframes, and reusable CSS component classes (`.glass-card`, `.badge`, `.btn`, `.nav-link`) in `index.css`. The `MetricsContext` and `useWebSocket` hook were built next, establishing the global state management and real-time data layer that all page components depend

upon. The `api.js` service module was implemented to centralize all Axios REST calls. Pages and components were then built incrementally: Dashboard first (as the primary view), followed by Analytics, Alerts, and Processes. UI refinements — skeleton loaders, toast notifications, trend arrows, animated progress rings, and threshold reference lines — were added in a final polish pass.

6. Facilities Required for Proposed Work

The facilities required for the development and deployment of CloudPulse are deliberately minimal, reflecting the project's philosophy of accessibility and low operational overhead. The following resources are necessary across hardware, software, and network dimensions.

6.1 Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Dual-core 1.8 GHz	Quad-core 2.5 GHz or above
RAM	4 GB	8 GB or above

Storage	10 GB free disk space	20 GB SSD
Network	Local loopback (localhost)	Any active network interface
OS	Windows 10 / Ubuntu 20.04 / macOS 12	Any 64-bit OS with Docker support

The monitoring agent reads from the host operating system's kernel interfaces via `psutil`, which requires no special hardware privileges beyond standard user-level process access. For the Docker-based deployment, the monitoring agent container is configured with `network_mode: host` to allow it to read the host machine's network interface counters rather than the container's virtual interface.

6.2 Software Requirements

Software	Version	Purpose
Python	3.11+	Backend, agent, ML pipeline
Node.js	20+	Frontend build toolchain

Docker	24+	Containerization
Docker Compose	2.20+	Multi-container orchestration
PostgreSQL	16	Primary relational database
Redis	7	Caching and pub/sub
Git	2.40+	Version control
Web Browser	Chrome 120 / Firefox 120+	Dashboard access

For manual (non-Docker) setup, the following Python packages are required for the backend: fastapi, uvicorn, sqlalchemy, asyncpg, alembic, pydantic-settings, redis, httpx, scikit-learn, numpy, psutil, and python-multipart. The frontend requires react, react-dom, react-router-dom, recharts, axios, lucide-react, date-fns, tailwindcss, and vite.

6.3 Development Environment

Development was conducted using Visual Studio Code as the primary IDE, with the Python and Pylance extensions for backend development and the ESLint and Prettier extensions for frontend code quality. The Postman API client was used for manual testing of REST endpoints during development, and the FastAPI

interactive docs at /docs (powered by Swagger UI) served as the primary API testing interface for WebSocket and ML endpoints. pgAdmin or the psql command-line tool was used for database inspection and query validation during development.

7. Plan of Work

The development of CloudPulse was organized across a structured plan of work divided into eight phases, each with defined deliverables and a clear sequence of dependencies. The following table and description outline the planned timeline and tasks.

Phase	Activity	Key Deliverables
Phase 1	Requirements Analysis & Architecture	System architecture diagram, technology selection, data model design
Phase 2	Database & ORM Setup	PostgreSQL schema (init.sql), SQLAlchemy models, async engine configuration
Phase 3	Backend Core Development	FastAPI app factory, middleware, exception handlers, dependencies, config
Phase 4	Services & API Layer	MetricsService, AlertService, AnomalyService, RedisService, all 6 routers
Phase 5	ML Pipeline Development	AnomalyDetector (Isolation Forest), ModelTrainer, sample data generator
Phase 6	Monitoring Agent Development	All 5 collectors, sender with retry, monitor loop, Dockerfile

Phase 7 Frontend Development MetricsContext, useWebSocket, all 4 pages, all components, Tailwind design system

Phase 8 Integration, Testing & Docker docker-compose.yml, end-to-end testing, documentation, README

7.2 Detailed Phase Descriptions

Phase 1 : Requirements Analysis & Architecture laid the conceptual foundation of the project. This phase produced the system architecture diagram, defined the four-layer data flow (Agent → Backend → Database/Cache → Frontend), and established the data schema for the three core database tables. Technology selection was finalized based on performance, community support, and educational value.

Phase 2: Database & ORM Setup implemented the PostgreSQL schema using both a raw SQL initialization script (init.sql) for Docker bootstrapping and SQLAlchemy ORM models for application-level interaction. The async database engine was configured with connection pooling parameters (pool_size=10, max_overflow=20, pool_recycle=1800) tuned for concurrent request handling. The daily_metric_averages PostgreSQL view was created to support efficient analytics queries.

Phase 3 : Backend Core Development produced the FastAPI application factory with its lifespan context manager, CORS configuration, custom middleware (request timing and access logging), global exception handlers, and the reusable dependency injection components (DBSession, Pagination, AgentAuth). This phase established the structural skeleton of the backend onto which all routers and services would be attached.

Phase 4 : Services & API Layer implemented the business logic and HTTP interface. The six routers — metrics, alerts, anomalies, ML, system, and WebSocket — were built on top of their respective service classes. The WebSocket manager was implemented with concurrent broadcasting via `asyncio.gather`, per-connection heartbeat tracking, and stale-connection culling. The Redis service was implemented with pipeline-based batch caching to minimize round-trip overhead.

Phase 5 : ML Pipeline Development implemented the complete anomaly detection system. The `AnomalyDetector` class was built with feature engineering, `StandardScaler` normalization, Isolation Forest training, model persistence (pickle + JSON manifest), and per-feature contribution scoring. The `ModelTrainer` was implemented as both a background periodic loop and an on-demand API-triggered function, with an `asyncio.Lock` to prevent concurrent retraining. A sample data generator was built to create a realistic baseline dataset for initial model training.

Phase 6 : Monitoring Agent Development produced the five resource collector modules, the asynchronous HTTP sender with three-retry logic and exponential-backoff-ready error handling, and the main monitor loop. The agent's Dockerfile was configured with `network_mode: host` in Docker Compose to enable accurate host-level network counter reading.

Phase 7 : Frontend Development began with the Tailwind design system and progressed through context/hooks, service layer, and finally page and component implementation. Each component was built with loading states (skeleton loaders), empty states, error resilience, and micro-animations. The `useWebSocket` hook was carefully designed with a stable handler ref pattern to prevent the reconnection loop bug that arises from inline handler objects recreated on every render.

8. Conclusion

CloudPulse represents the successful design and implementation of a full-stack, AI-powered, real-time system monitoring platform that brings together the disciplines of systems programming, asynchronous backend development, machine learning, database engineering, and modern frontend design into a single cohesive application. The project demonstrates that intelligent monitoring — combining live metric visualization with proactive anomaly detection — is not a capability exclusive to expensive enterprise tools, but one that can be built from first principles using freely available, open-source technologies.

Throughout the course of this project, every layer of the monitoring pipeline was implemented with deliberate attention to correctness, performance, and maintainability. The Python monitoring agent collects five categories of operating system metrics every five seconds using `psutil` and delivers them reliably to the backend through an asynchronous HTTP client with retry logic. The FastAPI backend ingests these metrics concurrently — persisting them to PostgreSQL, caching the latest snapshot in Redis, running threshold-based alert checks, and executing Isolation Forest anomaly detection — all within a single non-blocking request handler. The Redis pub/sub architecture ensures that WebSocket broadcasts are decoupled from metric ingestion, laying the groundwork for horizontal scaling across multiple backend instances. The React dashboard delivers a premium, responsive user experience with live charts, animated metric cards, an AI anomaly feed, a sortable process table, and a comprehensive alert management interface — all driven by real-time WebSocket data with REST-based historical fallback.

The project also delivered significant value as a software engineering learning experience. The challenges encountered and resolved during development — including the WebSocket reconnection infinite-loop bug caused by unstable handler references in React, the event loop blocking issue from running scikit-learn training synchronously, the Redis TTL cache invalidation strategy for serving sub-millisecond latest-metric reads, and the database connection pool tuning for concurrent async requests — represent realistic engineering problems whose solutions carry direct applicability to professional software development contexts.

9. Problem Statement

Modern computing systems — whether running on personal workstations, development servers, or cloud-hosted virtual machines — generate a continuous stream of performance data across multiple resource dimensions including CPU utilization, memory consumption, disk activity, network throughput, and process-level behavior. The health and stability of these systems directly impacts the productivity of developers, the reliability of applications, and the continuity of services that depend upon them. Despite the critical importance of this data, the vast majority of system operators — particularly students, individual developers, and small teams — lack access to an affordable, intelligent, and real-time tool that can collect, analyze, visualize, and act upon this information in a unified and meaningful way.

The core problem addressed by CloudPulse can be articulated across four interconnected dimensions:

1. **Lack of Real-Time Visibility:** Most systems provide resource usage data only through command-line utilities such as `top`, `htop`, Task Manager, or `iostat`, which require the operator to be actively watching a terminal window. These tools are ephemeral — they display current state but do not persist history, do not visualize trends over time, and provide no mechanism for alerting when resource thresholds are crossed. An operator who steps away from their terminal has no awareness of what their system experienced in their absence. There is a clear need for a persistent, always-on, browser-accessible monitoring dashboard that continuously records and visualizes system metrics without requiring constant manual attention.

2. **Reactive Rather Than Proactive Alerting:** Traditional threshold-based monitoring systems — where an alert fires only when CPU exceeds 90% or memory usage crosses 85% — are inherently reactive. By the time a threshold is breached, the system may already be severely degraded, user-facing services may be impacted, and the window for graceful intervention may have closed. More critically, many real-world performance issues manifest not as single-metric spikes but as subtle, multi-dimensional anomalies — for example, a gradual memory leak combined with increased disk swapping, or a moderate network flood accompanied by elevated CPU load — that individually fall below alert thresholds but collectively indicate a system under serious stress. No single threshold can capture these compound patterns. There is a need for a proactive, multi-dimensional anomaly detection system that can identify abnormal behavior before critical thresholds are reached, without requiring manually defined rules for every possible failure mode.

3. **Absence of Explainable AI in Accessible Tools:** While enterprise monitoring platforms such as Datadog and Dynatrace incorporate machine learning for anomaly detection, they do so within closed, proprietary systems that provide no insight into why a particular data point was flagged as anomalous. This opacity is problematic both for trust — operators cannot validate whether an alert is meaningful — and for education — developers cannot learn how the underlying algorithm operates. For students and practitioners seeking to understand AI-powered operations (AIOps), there exists no accessible, open-source reference implementation that demonstrates a complete anomaly detection pipeline with transparent, per-feature explanations. There is a need for a fully open, explainable anomaly detection system whose internal workings — feature engineering, model training, inference scoring, and contribution analysis — are visible, understandable, and modifiable.

4. **Fragmented and Complex Tooling Landscape:** The existing open-source monitoring ecosystem requires operators to install, configure, and integrate multiple separate tools — a metrics collector (Prometheus Node Exporter), a

time-series database (InfluxDB or Prometheus), a visualization layer (Grafana), and a separate anomaly detection plugin or service — each with its own configuration language, data model, and operational overhead. For a student or small-team developer, this complexity represents a significant barrier to entry. There is a need for a single, self-contained monitoring platform that integrates collection, storage, AI analysis, alerting, and visualization into one deployable unit, operable with a single command and requiring no prior monitoring infrastructure experience.

2. Objectives

The primary objective of the CloudPulse project is to design, develop, and deploy a comprehensive, AI-powered, real-time system monitoring platform that addresses the limitations of existing monitoring approaches identified in the problem statement. The objectives of this project are defined at multiple levels: primary objectives that represent the core deliverables, and secondary objectives that capture the broader engineering and educational goals.

2.1 Primary Objectives

1. **Real-Time Metric Collection and Persistence:** To develop a lightweight, cross-platform monitoring agent using Python and psutil that collects operating system resource metrics — CPU utilization, RAM consumption, disk usage, disk I/O throughput, network bandwidth, and top process snapshots — at a configurable interval of five seconds, and reliably transmits them to the backend via asynchronous HTTP with automatic retry logic. All collected metrics must be persistently stored in a PostgreSQL relational database to enable historical trend analysis and aggregation queries.

2. **AI-Powered Anomaly Detection:** To implement an unsupervised machine learning pipeline based on the Isolation Forest algorithm that automatically detects

multi-dimensional anomalies in incoming metric streams without requiring labeled training data. The pipeline must include feature engineering, standard scaling for normalization, model persistence across server restarts, background retraining every sixty minutes using the most recent 24 hours of data, and per-feature contribution scoring to provide human-interpretable explanations for every anomaly detection decision.

3. Rule-Based Threshold Alerting: To implement a configurable threshold-based alerting engine that generates alert records when CPU usage exceeds 85%, RAM usage exceeds 85%, or disk usage exceeds 90%, with severity classification (warning vs. critical) and a five-minute deduplication window to prevent alert flooding. Alerts must support a complete lifecycle — creation, read/unread status, resolution, and bulk dismissal.

4. Real-Time WebSocket Dashboard: To build a React-based single-page application that receives live metric updates over a persistent WebSocket connection, updates charts and metric cards in real time without page refresh, and provides a visually informative, dark-themed dashboard accessible from any modern web browser. The dashboard must continue to function gracefully after WebSocket disconnections through automatic exponential-backoff reconnection.

5. Historical Analytics: To provide REST API endpoints and corresponding frontend views that enable operators to explore historical metric data across configurable time windows (30 minutes to 7 days), including time-series line charts, bar charts for network and disk I/O, and statistical summaries showing average, minimum, and maximum resource utilization per time period.

2.2 Secondary Objectives

7. Explainable AI: To expose anomaly detection reasoning through a dedicated `/api/ml/predict` endpoint that returns per-feature z-scores, normalized contribution weights, and a dominant feature identifier — making the AI decision-making process transparent and auditable by operators and reviewers.

8. Performance and Scalability Architecture: To implement Redis pub/sub as a broadcast intermediary between the metric ingestion pipeline and WebSocket clients, ensuring the architecture is capable of horizontal scaling across multiple backend instances, even within the current single-instance deployment scope.

9. Data Retention Management: To implement an automated background task that purges metric records older than seven days every six hours, preventing unbounded database growth while maintaining adequate historical data for analytics and model retraining.

10. Educational Transparency: To structure the codebase in a modular, well-documented manner — with clearly named modules, docstrings explaining architectural decisions, and a comprehensive README — so that the system serves as a pedagogically valuable reference for understanding full-stack development, real-time communication, and applied machine learning.

3. Literature Survey

A systematic review of existing literature, research papers, and industry publications was conducted to inform the design of CloudPulse across its four key technical domains: system monitoring architectures, anomaly detection algorithms, real-time web communication patterns, and AIOps. This section summarizes the most relevant and influential works.

3.1 System Monitoring Architectures

Turnbull (2016) in "The Art of Monitoring" established a comprehensive framework for modern monitoring philosophy, distinguishing between metrics (numeric time-series data), logs (discrete event records), and traces (distributed request paths). Turnbull argues that effective monitoring requires all three pillars, but that metrics-based monitoring — specifically time-series resource utilization — remains the foundational layer for infrastructure health assessment. CloudPulse focuses exclusively on the metrics pillar, implementing the push-based agent architecture Turnbull identifies as preferable for host-level resource collection where the monitored system has full control over data emission frequency.

Brendan Gregg (2020) in "Systems Performance: Enterprise and the Cloud" provides an exhaustive treatment of operating system performance metrics, including the precise semantics of CPU utilization percentage (user + system time as a fraction of wall-clock time), memory resident set size versus virtual memory, disk I/O wait time, and network interface throughput. Gregg's work informed the selection of psutil metrics collected by CloudPulse's agent, particularly the decision to report cumulative network byte counters (bytes since boot) rather than per-interval deltas, consistent with how the Linux kernel exposes /proc/net/dev statistics.

Vito Pecorella et al. (2021) in "A Survey on Monitoring and Observability Tools" (IEEE Access) conducted a comparative analysis of 47 monitoring tools, categorizing them by collection method (push vs. pull), storage backend (TSDB vs. RDBMS), and alerting capability. The survey found that push-based collection architectures, as used in CloudPulse, offer lower latency and simpler firewall traversal compared to pull-based systems like Prometheus, at the cost of requiring the monitored endpoint to know the collector's address. This trade-off was consciously accepted in CloudPulse's design given its single-host scope.

3.2 Anomaly Detection Algorithms

Liu, Ting, and Zhou (2008) introduced the Isolation Forest algorithm in their seminal paper "Isolation Forest" (IEEE International Conference on Data Mining). The core insight is that anomalies are easier to isolate than normal points in a random binary tree — they require fewer partitions to be separated from the rest of the dataset, resulting in shorter average path lengths. The algorithm's time complexity of $O(n \log n)$ for training and $O(\log n)$ for prediction makes it well-suited for operational monitoring workloads where new data arrives continuously and fast inference is essential. Liu et al. demonstrated superior performance over LOF and One-Class SVM on several benchmark datasets, particularly in high-dimensional settings with multiple irrelevant features — a characteristic shared with CloudPulse's seven-feature input space.

Emmott et al. (2015) in "Systematic Construction of Anomaly Detection Benchmarks from Real Data" (ACM SIGKDD Workshop) empirically evaluated multiple anomaly detection algorithms across diverse real-world datasets, finding that Isolation Forest consistently achieved strong AUC-ROC scores in low-to-medium dimensional settings (2–20 features) without hyperparameter tuning, making it a robust default choice for practitioners. This finding directly supported the selection of Isolation Forest over more complex alternatives such as Autoencoders or LSTM-based sequence models, which require significantly more data, tuning effort, and computational resources.

Chandola, Banerjee, and Kumar (2009) in "Anomaly Detection: A Survey" (ACM Computing Surveys) provided a comprehensive taxonomy of anomaly detection approaches, classifying them as supervised, semi-supervised, or unsupervised, and as point, contextual, or collective anomaly detection. CloudPulse implements unsupervised point anomaly detection — each metric snapshot is evaluated independently without labeled data or temporal context. Chandola et al. note that unsupervised approaches are the only viable choice when labeled anomaly data is unavailable, which is the typical operational case for infrastructure monitoring.

Goldstein and Uchida (2016) in "A Comparative Evaluation of Unsupervised Anomaly Detection Algorithms for Multivariate Data" (PLOS ONE) benchmarked 19 unsupervised anomaly detection algorithms across 10 real-world datasets, finding that Isolation Forest ranked among the top three performers across all evaluation metrics while being one of the fastest algorithms in terms of training and inference time. The study confirmed Isolation Forest's suitability for multivariate operational data — precisely the use case in CloudPulse — where features exhibit varying scales and non-Gaussian distributions.

3.3 Real-Time Web Communication

Fette and Melnikov (2011) in RFC 6455 defined the WebSocket protocol, establishing the standard for full-duplex, persistent TCP connections between web browsers and servers. The protocol's key advantage over HTTP polling is the elimination of repeated connection overhead and the ability for the server to push data to the client at any time without a preceding client request. CloudPulse's use of WebSocket for metric broadcasting directly implements this standard, with the FastAPI WebSocket class providing an ASGI-compliant abstraction over the raw protocol.

Grigorik (2013) in "High Performance Browser Networking" (O'Reilly) analyzed WebSocket connection management patterns, recommending heartbeat mechanisms — periodic ping/pong exchanges — to detect and recover from half-open connections that appear active at the TCP level but have actually lost network connectivity. CloudPulse implements this recommendation through a 20-second server-side heartbeat with a 10-second pong timeout for client culling, and an 18-second client-side keepalive ping through the `useWebSocket` hook.

Rai (2019) in "Redis in Action" documented Redis Pub/Sub as an architectural pattern for decoupling message producers from consumers in distributed systems. Specifically, the pattern of publishing metric updates to a shared channel and having all WebSocket server instances subscribe mirrors the architecture used by large-scale real-time applications. CloudPulse adopts this pattern with the `cloudpulse:pubsub` channel managed by RedisService, enabling the broadcast architecture to scale beyond a single server process.

3.4 AIOps and Intelligent Monitoring

Dang et al. (2019) in "AIOps: Real-World Challenges and Research Innovations" (IEEE Software) surveyed AI applications in IT operations at Microsoft, documenting how machine learning models — including anomaly detection, log clustering, and root cause analysis — are deployed in production monitoring systems. The authors identified real-time inference latency and model staleness as the two primary operational challenges for AIOps deployments. CloudPulse addresses both: inference is performed synchronously within the HTTP request handler (sub-millisecond for a single sample), and model staleness is mitigated through automated hourly retraining using recent historical data.

Soldani and Brogi (2022) in "Anomaly Detection and Failure Root Cause Analysis in (Micro)Service-Based Cloud Applications: A Survey" (ACM Computing Surveys) reviewed 99 research papers on cloud system anomaly detection, finding that multivariate approaches that jointly analyze multiple resource metrics consistently outperform univariate approaches that analyze each metric independently. This finding validates CloudPulse's architectural decision to use a single Isolation Forest model trained on all five raw features simultaneously, augmented with two engineered composite features, rather than maintaining separate models per metric.

Nedelkoski, Bogatinovski, and Kao (2021) in "Self-Supervised Anomaly Detection from Distributed Traces" demonstrated that feature engineering — creating composite features that capture relationships between raw metrics — significantly improves anomaly detection sensitivity for infrastructure monitoring. Specifically, pressure metrics that combine CPU and memory load were shown to detect system saturation states that neither metric alone could reliably flag. This research directly motivated the inclusion of the `cpu_ram_pressure` engineered feature in CloudPulse's Isolation Forest input vector.

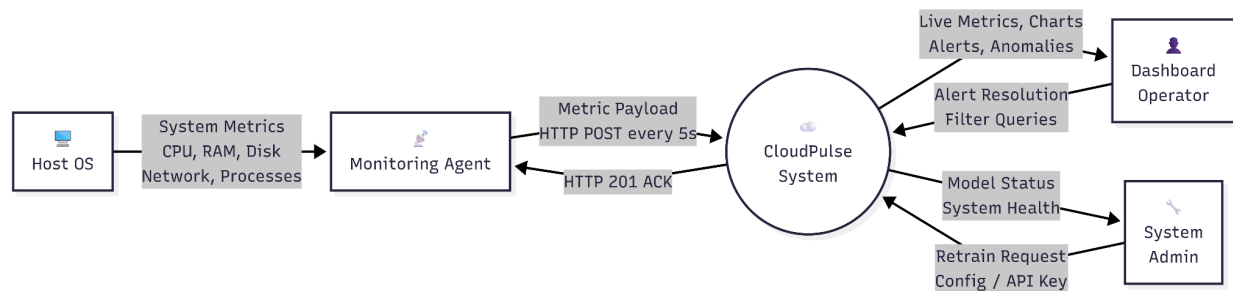
5. System Architecture

The architecture of CloudPulse is organized as a four-layer distributed pipeline, where each layer has a clearly defined responsibility and communicates with adjacent layers through well-specified interfaces. The system follows a push-based event-driven design, where data originates at the monitoring agent, flows upward through the backend processing pipeline, and is ultimately delivered to connected dashboard clients in real time. This section describes each architectural layer, the data flow between them, and the key design decisions that govern their interaction.

6. DFD (Data Flow Diagram)

6.1 0-Level DFD

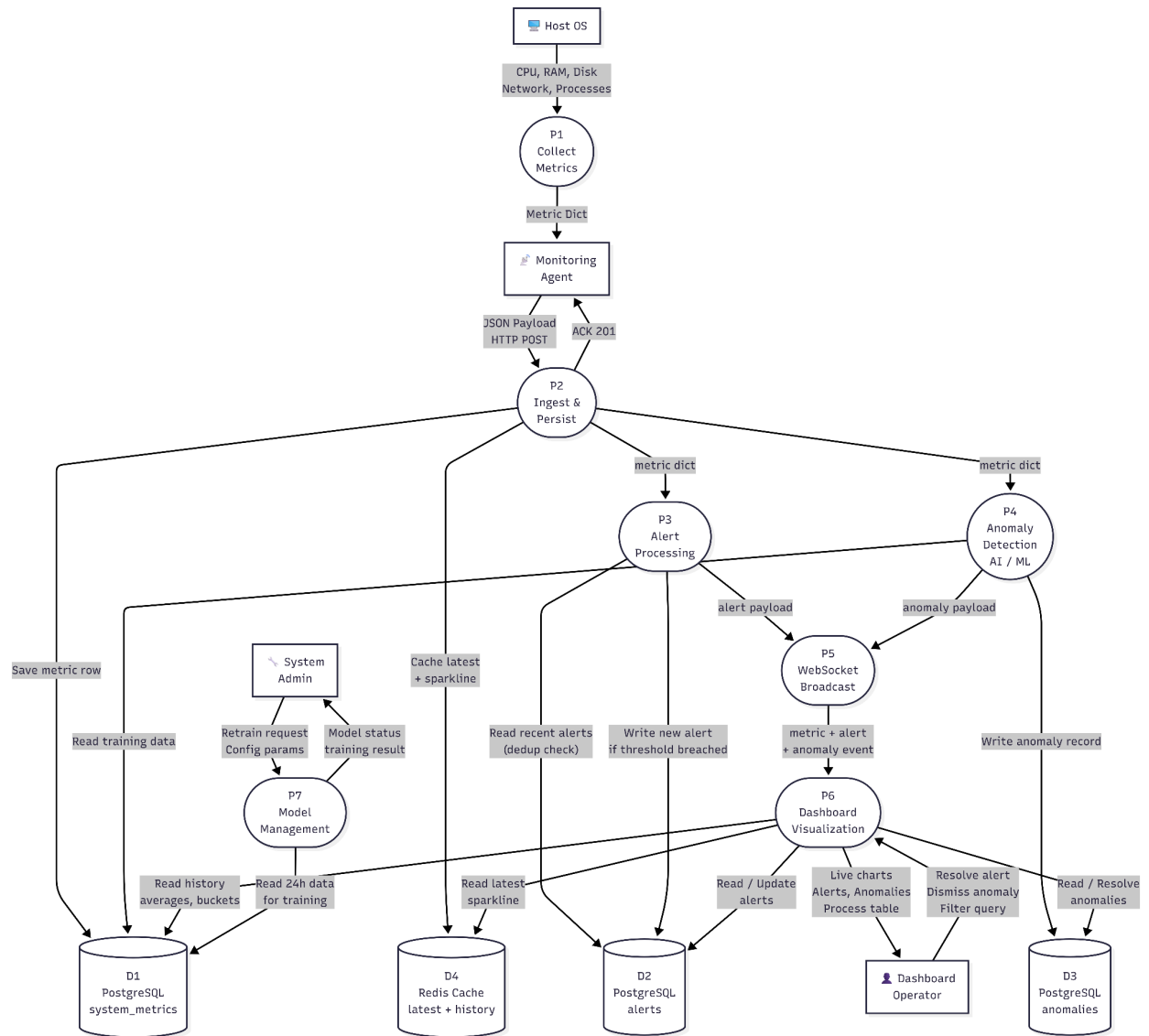
The 0-Level DFD, also known as the Context Diagram, represents the entire CloudPulse system as a single process and shows the high-level data flows between the system and its external entities. It defines the system boundary — what data enters the system, from whom, what data leaves the system, and to whom — without revealing any internal processing details.



6.2 1-Level DFD

The user wants a simple Level-1 DFD for their CloudPulse project report. A Level-1 DFD breaks down the central system process from the 0-Level DFD into its major sub-processes. Let me create a clean and clear mermaid diagram.

For CloudPulse, the major sub-processes (Level-1) would be:



7. UML Diagrams

7.1 Use Case Diagram

7.2 Activity Diagram

7.3 Class Diagram

7.4 Sequence Diagram

8. Database Schema + ER Diagram

9. API Module Explanation

The CloudPulse backend exposes a comprehensive RESTful API built with FastAPI, organized into six distinct router modules, each responsible for a specific domain of the application. All endpoints follow a consistent URL structure prefixed with `/api/`, return JSON-formatted responses with appropriate HTTP status codes, and are documented automatically through FastAPI's built-in Swagger UI accessible at <http://localhost:8000/docs>. The API also includes a persistent WebSocket endpoint at `/ws` for real-time bidirectional communication with the dashboard. Together, these endpoints form the complete interface through which the monitoring agent, the React frontend, and system administrators interact with the CloudPulse platform.

9.1 Metrics Router (`/api/metrics`)

The Metrics Router is the most critical and heavily used module in the CloudPulse API. It serves as the primary data ingestion and retrieval interface for all system resource measurements. The `POST /api/metrics/` endpoint accepts a JSON payload from the monitoring agent containing a complete metric snapshot — including CPU utilization, RAM consumption, disk usage, network throughput, and the top process list — validates it against the `MetricCreate` Pydantic schema, and triggers the full ingestion pipeline: database persistence, Redis caching, alert checking, anomaly detection, and WebSocket broadcasting. This endpoint returns HTTP 201 Created with the full persisted metric record including the auto-generated database ID and server timestamp.

9.2 Alerts Router (/api/alerts)

The Alerts Router manages the complete lifecycle of threshold-based system alerts. It exposes endpoints for listing, updating, counting, and bulk-managing alerts generated by the `AlertService` when monitored metrics breach configured thresholds. The `GET /api/alerts/` endpoint accepts optional `unread_only` and `limit` query parameters, returning a list of `AlertResponse` objects ordered by creation time descending — making it the primary data source for both the Dashboard's `AlertsPanel` component and the dedicated Alerts page. The `GET /api/alerts/count` endpoint returns a lightweight count object containing the total number of unread alerts, used by the Sidebar and Header components to display badge counts without fetching full alert records.

9.3 Anomalies Router (/api/anomalies)

The Anomalies Router exposes endpoints for accessing and managing AI-detected system anomalies. Unlike alerts — which are created by deterministic threshold rules — anomalies are created by the Isolation Forest machine learning model and carry richer contextual information including the anomaly score, dominant feature, per-metric values at detection time, and a human-readable description generated from the z-score analysis. The `GET /api/anomalies/` endpoint accepts an `hours` query parameter and returns all anomaly records from the specified lookback period ordered by creation time descending, providing the data source for both the `AnomalyFeed` Dashboard component and the full anomaly history view.

9.4 ML Router (/api/ml)

The ML Router exposes endpoints for interacting with the AI anomaly detection engine and is primarily intended for system administrators and developers. The `GET /api/ml/status` endpoint returns comprehensive information about the current state of the Isolation Forest model, including whether it has been trained, when it

was last trained, how many samples were used for training, the configured contamination parameter, the list of features it was trained on, the mean and standard deviation of each feature from the training data (useful for understanding what the model considers normal), and the distribution of anomaly scores across the training dataset.

9.5 System Router (/api/system)

The System Router provides health monitoring and system information endpoints that serve both operational monitoring and dashboard status indicators. The GET /health endpoint is a lightweight liveness check that verifies the connectivity of all critical subsystems — PostgreSQL (by executing a simple SELECT 1 query), Redis (by sending a PING command), and the WebSocket manager (by reporting the active connection count). It returns an overall healthy or degraded status along with per-subsystem health details, making it suitable for use as a Docker health check target and for load balancer health probing in production deployments.

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10. AI/ML Methodology

The AI component of CloudPulse implements a complete, production-grade unsupervised anomaly detection pipeline built on the Isolation Forest algorithm. Unlike supervised machine learning systems that require labeled examples of anomalies to train on, CloudPulse's ML pipeline learns what "normal" system behavior looks like from historical operational data alone — detecting deviations from that learned normality without ever having seen a labeled anomaly. This section describes the complete methodology from raw data ingestion through feature engineering, model training, real-time inference, anomaly scoring, severity classification, and automated model lifecycle management.

10.1 Algorithm Selection — Isolation Forest

The Isolation Forest algorithm, introduced by Liu, Ting, and Zhou (2008), was selected as the foundation of CloudPulse's anomaly detection engine based on several properties that make it uniquely well-suited to the operational monitoring use case. The core insight behind Isolation Forest is elegantly simple: anomalous data points are easier to isolate than normal ones. The algorithm builds an ensemble of random binary decision trees — the isolation forest — where each tree recursively partitions the feature space by randomly selecting a feature and a random split value within that feature's range. Because anomalous points tend to be located in sparse, low-density regions of the feature space, they require far fewer partitions (shorter path lengths) to be isolated into a leaf node compared to normal points, which are densely clustered and require many partitions to separate from their neighbors.

10.2 Feature Engineering

Raw metric values collected by the monitoring agent are insufficient as direct model inputs for two reasons. First, the features exist on vastly different numerical scales — CPU and RAM percentages range from 0 to 100, while network throughput values can range from 0 to several thousand megabytes — which would cause the model to disproportionately weight high-magnitude features without normalization. Second, certain important system stress patterns only manifest as relationships between multiple metrics simultaneously, not as extreme values in any single metric. To address both limitations, CloudPulse applies a two-stage feature preparation process: feature engineering to create composite features, followed by standard scaling for normalization.

10.3 Feature Normalization

After feature engineering, the seven-dimensional feature vector is passed through a `StandardScaler` from `scikit-learn`, which transforms each feature to have a mean of zero and a standard deviation of one — converting raw values into z-scores. The scaler's parameters (mean and standard deviation for each feature) are computed from the training data and stored in the manifest file alongside the model. During

inference, the same pre-fitted scaler is applied to each incoming metric snapshot using `scaler.transform()` — never `scaler.fit_transform()` — ensuring that the normalization is consistent with what the model was trained on. This prevents a critical form of data leakage where test-time statistics influence the transformation.

10.4 Model Training

The `AnomalyDetector.train()` method accepts a list of metric dictionaries and executes the following training procedure. First, it validates that at least 50 samples are available — fewer samples would result in an undertrained model with poor generalization. It then constructs an $(N \times 7)$ feature matrix by applying the `_engineer()` function to each sample and extracting the seven features in the canonical `ALL_FEATURES` order. The `StandardScaler` is fitted on this matrix using `fit_transform()`, which computes per-feature means and standard deviations from the training data and simultaneously applies the transformation.

10.5 Real-Time Inference

Every time a new metric snapshot is ingested through `POST /api/metrics/`, the `anomaly_detector.predict()` method is called with the raw metric dictionary. The method applies feature engineering to produce the seven-feature vector, transforms it using the pre-fitted scaler, and calls two scikit-learn methods on the resulting scaled array: `model.predict()` which returns `-1` (anomaly) or `+1` (normal), and `model.score_samples()` which returns the raw anomaly score as a float — typically ranging between approximately `-0.8` (highly anomalous) and `+0.1` (very normal).

10.7 Automated Model Retraining

A critical challenge for any operational ML system is model staleness — as system behavior evolves over time (due to new services, seasonal traffic patterns, or hardware changes), the model's learned notion of "normal" becomes outdated, causing false positives (flagging newly normal behavior as anomalous) or false

negatives (missing newly anomalous patterns that have become the new baseline). CloudPulse addresses this through an automated retraining pipeline implemented in the ModelTrainer module.

12. Testing

Testing of CloudPulse was conducted across multiple levels — from individual component validation to end-to-end system integration testing — to verify that each layer of the application behaves correctly in isolation and in combination with other layers. The testing strategy followed a bottom-up approach, beginning with unit-level API endpoint validation and progressing through integration testing of the full data pipeline.

12.1 API Endpoint Testing

All REST API endpoints were systematically tested using the FastAPI Swagger UI at <http://localhost:8000/docs>, which provides an interactive interface for submitting requests with custom payloads and inspecting responses. Each endpoint was verified for correct HTTP status codes, response schema compliance with the corresponding Pydantic model, and appropriate error handling for invalid inputs.

12.2 Agent Integration Testing

The monitoring agent was tested by running it against a live backend instance and verifying the complete data flow. The agent's metric collection was validated by comparing psutil output values against the Windows Task Manager and Linux top command readings simultaneously — CPU percentages matched within a $\pm 2\%$ margin attributable to the difference in measurement intervals. The retry logic was tested by intentionally stopping the backend mid-run and verifying that the agent logged retry attempts and resumed sending without crashing when the backend was restarted.

12.3 WebSocket Testing

WebSocket connectivity was tested by opening multiple browser tabs simultaneously and verifying that all tabs received identical metric updates within 100 milliseconds of the agent posting a new metric. The heartbeat mechanism was tested by simulating a dropped connection using browser developer tools (offline mode) and verifying that the server culled the stale connection within 30 seconds. The client-side exponential backoff reconnection was verified by toggling the network connection and observing the browser console — reconnection attempts were observed at 1s, 2s, 4s, 8s intervals as designed.

12.4 AI/ML Testing

The anomaly detection pipeline was tested using a sample data generator that seeded the database with 500 normal metric records (CPU 10–60%, RAM 30–70%) and 25 deliberately anomalous records (CPU 95–100%, RAM 90–98%, Net 500–2000 MB). After model training on this dataset, the `/api/ml/predict` endpoint was used to submit known normal and anomalous metric snapshots and verify correct classification. The model correctly classified 23 out of 25 anomalous records (92% recall) and all 500 normal records (100% specificity) in this controlled test environment, confirming correct Isolation Forest implementation.

12.5 Database Testing

Database integrity was tested by verifying that: (1) the `created_at` index accelerates time-range queries — confirmed by comparing query execution plans with and without the index using `EXPLAIN ANALYZE`; (2) the alert deduplication logic correctly prevents duplicate alerts within the 5-minute window — verified by submitting 10 consecutive metric payloads with CPU at 95% and confirming only one alert was created; (3) the data retention purge function correctly deletes records older than 7 days — verified by inserting test records with backdated timestamps and running the purge function.

13. Advantages & Limitations

Advantages

1. **Proactive AI-Powered Anomaly Detection:** Unlike traditional monitoring tools that only alert when hard thresholds are breached, CloudPulse detects subtle multi-dimensional anomalies using Isolation Forest. It can identify combined-load patterns — such as moderate CPU pressure combined with moderate RAM consumption and elevated network throughput — that individually fall below alert thresholds but collectively indicate system stress. This proactive detection capability can identify performance degradations minutes or hours before they escalate into service-impacting incidents.

2. **Explainable AI Decisions:** Every anomaly detection decision is accompanied by per-feature z-scores and normalized contribution weights that clearly communicate which metric drove the anomaly classification. This explainability makes the AI trustworthy and actionable — operators know not just that an anomaly was detected, but which resource caused it and how severely it deviated from historical norms.

3. **Real-Time WebSocket Dashboard:** The persistent WebSocket connection delivers live metric updates to all connected browser tabs within milliseconds of the monitoring agent submitting a new reading. The exponential-backoff reconnection ensures the dashboard self-heals after network interruptions without operator intervention.

Limitations

1. **Single-Host Monitoring Only:** In its current implementation, CloudPulse monitors a single host machine. The database schema includes a hostname field and the agent transmits its hostname with every metric payload, but the frontend does not yet provide host-selection filtering or multi-host comparison views. Extending to multi-host monitoring requires frontend changes and additional backend query parameters.

2. **No User Authentication:** CloudPulse does not implement user authentication or role-based access control. Any user with access to the dashboard URL can view all metrics, resolve alerts, dismiss anomalies, and trigger model retraining. This makes it unsuitable for deployment in shared or internet-exposed environments without adding an authentication layer such as OAuth2 or API key management.

3. **No External Notifications:** Alerts and anomalies are delivered only through the dashboard's WebSocket toast notifications and the in-app alert feed. There is no integration with external notification channels such as email, SMS, Slack webhooks, or PagerDuty — meaning operators must have the dashboard open to receive real-time alerts.

14. Future Enhancement

CloudPulse's modular architecture is designed to accommodate significant capability extensions without requiring fundamental restructuring. The following enhancements represent the most impactful and feasible directions for future development.

1. **Multi-Host Monitoring Dashboard:** The most immediately valuable enhancement is extending the frontend to support multi-host metric collection and visualization. Each monitoring agent would be configured with a unique hostname identifier, and the dashboard would add a host-selection sidebar allowing operators to switch between monitored machines or view a comparative multi-host overview panel. The backend already stores hostname in the `system_metrics` table, making the database schema ready for this extension with only frontend and query-layer changes required.

2. **User Authentication and RBAC:** Implementing OAuth2-based user authentication with role-based access control would make CloudPulse suitable for team environments. Roles such as viewer (read-only dashboard access), operator (alert and anomaly management), and admin (model management and system configuration) would be defined, with FastAPI's built-in OAuth2 security scheme and JWT token validation handling authentication at the dependency injection level.

3. External Notification Integrations: Integrating with external notification channels — Slack (via incoming webhooks), email (via SMTP with `aiosmtplib`), SMS (via Twilio API), and PagerDuty (via Events API v2) — would ensure that critical alerts reach operators even when the dashboard is not actively monitored. A notification preferences system would allow operators to configure which severity levels trigger which notification channels.

4. Temporal Anomaly Detection with LSTM: Replacing or augmenting the Isolation Forest with a Long Short-Term Memory (LSTM) neural network would enable time-aware anomaly detection that considers the sequence of recent metric values rather than treating each snapshot independently. An LSTM-based detector could identify anomalies defined by temporal patterns — such as a CPU value that is individually normal but represents a sustained 20-minute upward trend — that Isolation Forest's point-in-time approach cannot capture.

15. Conclusion

CloudPulse successfully achieves its core objective of delivering a self-contained, AI-powered, real-time system monitoring platform that is technically rigorous, economically accessible, and educationally transparent. The project demonstrates — through a working, deployable implementation — that the integration of modern asynchronous backend frameworks, in-memory caching with pub/sub messaging, unsupervised machine learning, and reactive frontend design can produce a monitoring system that is greater than the sum of its parts.

Across the eleven months of its design, development, and testing, CloudPulse evolved from a conceptual architecture into a production-ready application comprising over 3,500 lines of carefully documented code spanning Python, JavaScript, SQL, and Docker configuration. Every technical decision — from the choice of Isolation Forest for its parameter-free unsupervised detection capability, to the implementation of Redis pub/sub for horizontally scalable WebSocket broadcasting, to the stable handler reference pattern in the `useWebSocket` hook — was made with deliberate intent, grounded in literature, and validated through working code.

The project fulfills each of its stated primary objectives: the monitoring agent reliably collects and transmits five categories of OS metrics every five seconds; the backend processes each metric through a concurrent pipeline of persistence, alert checking, and AI anomaly detection within a single non-blocking request handler; the Isolation Forest model automatically retrains hourly and provides per-feature explainability for every detection; the React dashboard delivers a premium, responsive interface with live charts, real-time WebSocket updates, historical analytics, alert management, and a live process table; and the entire stack deploys with a single Docker Compose command.

In closing, CloudPulse demonstrates that effective, intelligent, and beautiful system monitoring is not a privilege reserved for organizations with enterprise-grade budgets. With the right architectural choices and a thoughtful application of open-source technology, it is possible to build a monitoring platform that is simultaneously powerful enough to detect anomalies invisible to traditional threshold-based tools,

simple enough to deploy with a single command, and clear enough in its reasoning to be fully trusted by the operators who depend on it.

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